

1. E-commerce Customer Behavior & Revenue Optimization

Description: Analyze customer purchase behavior to identify high-value users, detect churn risk, and propose strategies to increase revenue.

Tech Stack: Python (Pandas, Matplotlib/Seaborn), SQL, Power BI/Tableau, Excel

Key Steps:

- 1 Load and clean transaction data
- 2 Perform customer segmentation (RFM or clustering)
- 3 Analyze revenue contribution by segment
- 4 Identify churn-risk customers
- 5 Build an interactive dashboard
- 6 Write business recommendations

2. A/B Testing & Product Decision Analysis

Description: Evaluate whether a new product feature improves performance using statistical testing.

Tech Stack: Python (SciPy, Pandas), Excel, basic statistics

Key Steps:

- 1 Define hypothesis and success metric
- 2 Prepare control vs treatment dataset
- 3 Perform statistical test (t-test or proportion test)
- 4 Interpret p-value and confidence intervals
- 5 Draw conclusion (launch or not)
- 6 Document decision reasoning

3. Job Market Analysis Dashboard

Description: Analyze job listings to understand in-demand skills, salary trends, and hiring patterns.

Tech Stack: Python (Pandas, BeautifulSoup optional), SQL, Power BI/Tableau

Key Steps:

- 1 Collect or load job market data
- 2 Clean and structure dataset
- 3 Analyze skills demand and salary distribution
- 4 Identify key hiring trends
- 5 Build a dashboard
- 6 Summarize insights in a short report